

# SEC P vs NP log 2026-05-20

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-20 00:03 UTC

## SEC P vs NP — daily log 2026-05-20

1 cycles recorded

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### Algebraic Degree of Tseitin Variety and Resolution Width

- **Verdict:** INCONCLUSIVE
- **Bridge:** ALGEBRAIC\_GEOMETRY  $\times$  RESOLUTION\_WIDTH
- **Recorded:** 2026-05-20 00:03 UTC
- **Entry ID:** 0f0cb06a6e41

#### Statement

For a Tseitin formula derived from a  $d$ -regular expander graph  $G$  with  $n$  vertices, the resolution width required to refute it is at least the algebraic degree of the variety defined by the system of polynomial equations corresponding to the clauses of the formula. This degree is computed as the minimal number of variables in a non-degenerate polynomial parametrization of the variety.

#### Rationale

Tseitin formulas on expanders are known to require super-polynomial resolution refutations. By linking their complexity to the algebraic geometry of the underlying variety, we explore whether the intrinsic geometric complexity (measured by the algebraic degree) directly correlates with the proof complexity. This bridges the gap between combinatorial proof systems and algebraic invariants of the problem structure.

#### Novelty

- Judge: NOVEL over 0 arXiv hits

#### Empirical Test

- exit code: 1, elapsed: 0.09s

TRIAL: {'metric\_name': 'resolution\_width', 'metric\_value': 1, 'instances\_tested': 1, 'conjecture\_hol

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_12718bfc.py", line 116, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_12718bfc.py", line 71, in run_trial
    G = random_d_regular_expander_graph(n, d)
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnr_sandbox/test_12718bfc.py", line 87, in random_d_regu
    raise ValueError("Invalid parameters for expander graph")
ValueError: Invalid parameters for expander graph
```

## Judge reasoning

Test crashed during parameter validation, preventing reliable evaluation of the conjecture | next: Verify validity of d-regular expander graph parameters and retry experiment